CAMERA CALIBRATION

Project 1 – Computer Vision and Pattern Recognition

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1. Problem Statement

Camera calibration is the process of estimating the intrinsic and extrinsic parameters of a camera from a set of observations. Intrinsic parameters (focal lengths, principal point, and lens skew) describe the internal optics of the camera and are collected in the calibration matrix K. Extrinsic parameters (rotation matrix R and translation vector t) describe the pose of the camera with respect to a reference coordinate system, and differ for each acquired image.

This project implements the Zhang calibration procedure [Zhang, 2002] from scratch, using a planar chessboard pattern as the calibration target. The main tasks are:

1. Calibrate using the Zhang procedure [Zhang, 2002], i.e., find the intrinsic parameters K and, for each image, the pair of R, t(extrinsic);
2. Choose one of the calibration images and compute the total reprojection error for all the grid points (adding a figure with the reprojected points);
3. Superimpose an object (for instance, a cylinder as in Fig. 1), to the calibration plane, in 25 images of your choice and check the correctness of the results by visual inspection;
4. Optionally, plot the standard deviation of the principal point (entries u_0 and v_0 of the calibration matrix K) as a function of the number of images used for calibration.

2. Camera Model

The pinhole camera model is adopted. A 3D point $\mathrm{M} = (X, Y, Z)^T$ is projected onto the image plane as:

$$\lambda \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \mathrm{K}\,[R\,|\mathrm{t}] \begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix}$$

where K is the intrinsic calibration matrix and (R, t) are the extrinsic parameters. The intrinsic matrix is defined as:

$$\mathrm{K} = \begin{bmatrix} \alpha_x & s & u_0 \\ 0 & \alpha_y & v_0 \\ 0 & 0 & 1 \end{bmatrix}$$

where α_x , α_y are the focal lengths in pixels, s is the skew parameter, and $(u_0,\ v_0)$ is the principal point.

3. Description of the Approach

3.1 Calibration Target

A planar chessboard pattern with 8x11 inner corners is used as the calibration target. Its real-world coordinates are placed on the Z = 0 plane, with a square size of 11 mm, the mapping between the world plane and the image plane is described by a homography:

$$\lambda \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \mathrm{H} \begin{bmatrix} X \\ Y \\ Z \end{bmatrix}$$

The world coordinates of each corner are computed using np.unravel_index to map the flat corner index to a 2D grid position, then multiplied by the square size to obtain millimetre coordinates.

3.2 Corner Detection

For each image, OpenCV's findChessboardCorners estimates the inner corners of the pattern. Detected corners are refined to sub-pixel accuracy with cornerSubPix, using a search window of 5x5 pixels and a stopping criterion of 100 iterations or a displacement below 0.001 pixels.

3.3 Homography Estimation

The homography H is estimated using a Direct Linear Transform (DLT) formulation. Each point correspondence ($m_i \leftrightarrow m_i'$) provides two linear equations:

$$\begin{bmatrix} m^T & 0^T & -um^T \\ 0^T & m^T & -vm^T \end{bmatrix} \begin{bmatrix} h_1 \\ h_2 \\ h_3 \end{bmatrix} = 0$$

where $\mathrm{m} = (X, Y, 1)^T$. With N correspondences we get a tall matrix A of shape $(2N \times 9)$. The solution h is the right singular vector of A corresponding to the smallest singular value, obtained via SVD. H is then recovered by reshaping h into a 3x3 matrix and normalising so that $\mathrm{H}[2,2] = 1$.

To visually verify the correctness of the estimated homographies, a planar rectangle defined in the checkerboard coordinate system (bottom-left corner at (11, 33) mm, width 99 mm, height 44 mm) was projected onto the calibration images using H. The projected rectangle aligns accurately with the checkerboard plane, confirming the validity of the homography estimation (Figure 1).

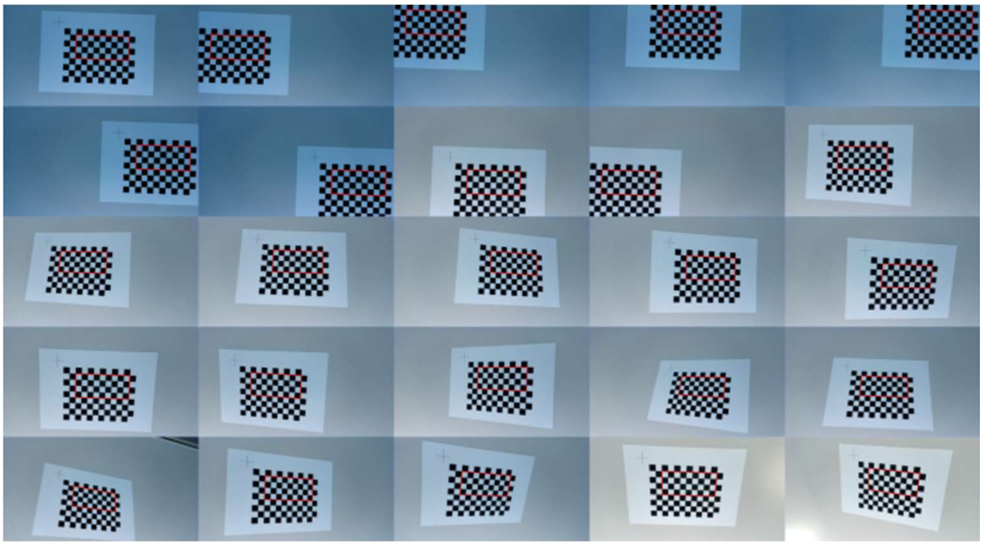

Figure 1

3.4 Intrinsic Calibration (Zhang’s Method)

After estimating the homographies H_i (one per image), the first step of Zhang's procedure is to estimate the symmetric positive definite matrix:

$$\mathrm{B} = K^T K^{-1}$$

Zhang's method uses the fact that, for a planar target, the homography columns $\mathrm{h1}$ and $\mathrm{h2}$ are related to the calibration matrix K through:

1. $h_1^T \mathrm{B} h_2 = 0$ (ortogonality constraint)
2. $h_1^T \mathrm{B} h_1 - h_2^T \mathrm{B} h_2 = 0$ (equal norm constraint)

Introducing the vector v_{ij} associated to a homography H = [h1 h2 h3]:

$$v_{ij} = \begin{bmatrix} h_{1i}h_{1j} \\ h_{1i}h_{2j} + h_{2i}h1j \\ h_{2i}h_{2j} \\ h_{3i}h_{1j} + h_{1i}h_{3j} \\ h_{3i}h_{2j} + h_{2i}h_{3j} \\ h_{3i}h3_j \end{bmatrix}, \mathrm{i,j} \in 1{,}2{,}3$$

each homography contributes two linear constraints on b $= [B11,\ B12,\ B22,\ B13,\ B23,\ B33]^T$:

$$\begin{cases} v_{12}^T B h_2 = 0 \\ (v_{11} - v_{22})^T = 0 \end{cases}$$

Stacking the equations for all M images yields the homogeneous linear system Vb $= 0$, where V has shape (2M x 6). The solution b is the last row of V^T from SVD. Since the SVD normalises singular vectors to unit norm and chooses their sign arbitrarily, and since B must be positive definite (required for Cholesky factorization), we enforce B11 > 0 by negating b if necessary.

The symmetric matrix B is then reconstructed as:

$$\mathrm{B} = \begin{bmatrix} b_1 & b_2 & b_4 \\ b_2 & b_3 & b_5 \\ b_4 & b_5 & b_6 \end{bmatrix}$$

Since B is symmetric and positive definite, it admits a unique Cholesky factorization $\mathrm{B} = \mathrm{L}\,L^T$. Comparing with $\mathrm{B} = (KK^T) = (K^{-1})^T K^{-1}$ gives $K^{-1} = L^T \Longrightarrow K = (L^T)^{-1}$. The matrix is normalised so that $\mathrm{K}[2{,}2] = 1$.

3.5 Extrinsic Parameters

For each image, the extrinsic parameters R and t are computed from K and $\mathrm{H} = [h1h2h3]$ as:

$$\lambda = \frac{1}{||K^{-1}h_1||}$$

$$r_1 = \lambda K^{-1} h_1$$

$$r_2 = \lambda K^{-1} h_2$$

$$r_3 = r_1 \times r_2$$

$$\mathrm{t} = \lambda K^{-1} h_3$$

Due to measurement noise in the detected corners and numerical inaccuracies in the homography estimation, the raw matrix $R_{approx} = [r1r2r3]$ may not exactly satisfy the constraints of SO(3) (Special Orthogonal Group in 3 dimensions). To obtain a valid rotation matrix, R_{approx} is projected onto the nearest element of SO(3) using SVD:

$$R_{approx} = \mathrm{USV}^T \Longrightarrow R = \mathrm{UV}^T$$

The orthonormality of R was verified by computing R^T R and det(R). After SVD orthonormalisation, the diagonal errors of R^T R - I is in the order of 10^{-16}, the maximum off-diagonal error is 1.20×10^{-17}, and $\det(R) = 0.9999999999999998 \approx 1$. This confirms that the final rotation matrix exactly satisfies the constraints of SO(3).

If $\det(R) = -1$ (a reflection rather than a rotation), the last column of U is negated, and t is flipped to ensure $\det(R) = +1$.

3.6 Reprojection Error

The image rgb_0.png was loaded and converted to grayscale. The inner corners of the checkerboard pattern (8x11 = 88 corners) were detected using findChessboardCorners and refined to sub-pixel accuracy using cornerSubPix with a 5x5 search window, stopping after 100 iterations or when displacement is below 0.001 pixels. For each detected corner, the corresponding real-world coordinates are computed on the plane Z = 0, with square size 11 mm. Since all world points lie on the calibration plane, the projection reduces to a planar homography $H_{pred} = K[r_1 \quad r_2 \quad t]$. Each world point $m_i = (X_i, Y_i, 1)^T$ is projected onto the image plane $\widetilde{m}_i = H_{pred} m_i$ and dehomogenised to get the predicted pixel coordinates $\hat{u}_i = \frac{\tilde{u}_i}{\widetilde{w}_i}, \hat{v}_i = \frac{\tilde{v}_i}{\widetilde{w}_i}$. The reprojection error for each point is the Euclidean distance between the predicted pixel coordinates $(\hat{u}_i, \hat{v}_i)$ and the measure image coordinates (u_i, v_i). The total reprojection error is defined as:

$$\epsilon(P) = \sum_{i=1}^{N} ((\hat{u}_i - u_i)^2 + (\hat{v}_i - v_i)^2)$$

The average squared reprojection error per point is calculated as: $\frac{\epsilon(P)}{88}$

The corresponding Root Mean Square Error (RMSE) is: $\sqrt{\frac{\epsilon(P)}{88}}$

3.7 Cylinder Augmentation

A maximum of M = 25 calibration images were selected (indices 0 to 24). The cylinder is defined in metric units (mm), consistently with the calibration. Its geometric parameters are:

- Base centre: $(x_0, y_0) = (40{,}50)\mathrm{mm}$
- Radius: $\mathrm{r} = 15\ \mathrm{mm}$
- Height: $\mathrm{h} = 80\ \mathrm{mm}$
- Base on the calibration plane $\mathrm{Z} = 0$; top at $\mathrm{Z} = 80\ \mathrm{mm}$

Both circles are sampled at $N_\theta = 40$ equally spaced angles. The sampled 3D points for base and top are:

$$m_{base_k} = (x_0 + r * \cos(\theta_k), y_0 + r * \sin(\theta_k), 0)^T$$

$$m_{to\ \ k} = (x_0 + r * \cos(\theta_k), y_0 + r * \sin(\theta_k), h)^T$$

$$\theta_k = \frac{2\pi k}{N_\theta}, \qquad \text{with } \mathrm{k} = 0, \dots, N_\theta - 1$$

To project the cylinder points into each image, the pinhole camera model is applied. Given K and the extrinsic parameters (R, t), the projection matrix is: $P = K\,[R \mid t]$ (3x4 matrix). Each 3D point $M_i = (X_i, Y_i, Z_i)^T$ is converted to homogeneous form $\widetilde{M}_i = (X_i, Y_i, Z_i, 1)^T$ and projected as:

$$\widetilde{m}_i = P\, \widetilde{M}_i = (\tilde{u}_i, \tilde{v}_i, \widetilde{w}_i)^T$$

where $(\tilde{u}_i, \tilde{v}_i, \widetilde{w}_i)$ are the homogeneous image coordinates. The final pixel coordinates are obtained by dehomogenisation:

$$\hat{u}_i = \frac{\tilde{u}_i}{\widetilde{w}_i}, \hat{v}_i = \frac{\tilde{v}_i}{\widetilde{w}_i}$$

This is done separately for the base and top circle, giving two outlines per image.

The cylinder is rendered by drawing the projected base circle (blue), top circle (red), and lateral edges (green) as line segments.

3.8 Influence of Number of Calibration Images

To study the stability of the intrinsic parameters as a function of the number of calibration images, all possible subsets of size ℓ (ℓ = 2, 3, ..., 20) were enumerated from a fixed set of M = 20 images. For each subset, K was re-estimated using Zhang's linear method. Subsets where B is not positive definite (Cholesky factorization fails) are discarded. The principal point coordinates $u_0 = \mathrm{K}[0,2]$ and $v_0 = \mathrm{K}[1,2]$ were tracked and the standard deviation was computed across all valid subsets for each value of ℓ.

4. Results

4.1 Estimated Intrinsic Matrix K and Extrinsic Parameters

Using all 81 successfully processed calibration images, the Zhang procedure converges to the following intrinsic matrix:

$$\mathrm{K} = \begin{bmatrix} 1010.00 & 1.69 & 630.35 \\ 0.00 & 1010.61 & 373.90 \\ 0.00 & 0.00 & 1.00 \end{bmatrix}$$

The focal lengths $\alpha_x \approx 1010.00\mathrm{px}$ and $\alpha_y \approx 1010.61\mathrm{px}$ are nearly equal, confirming that the camera sensor has approximately square pixels. The skew parameter $s \approx 1.69\mathrm{px}$ is small. The principal point $(c_x, c_y) \approx (630.35{,}373.90)\mathrm{px}$ is close to but not exactly at the image centre, reflecting a small but realistic lens misalignment.

Extrinsic parameters were successfully computed for all 81 calibration images. The determinant of $\det(R) \approx 0.999$ (machine precision: 0.9999999999999998), confirming a valid rotation. The translation vector indicates the camera was approximately 286 mm away from the calibration plane along the optical axis.

4.2 Reprojection Error

The reprojection error is computed on image rgb_0.png by projecting all 88 grid corners (8x11) through the estimated $\mathrm{K},\ \mathrm{R},\ \mathrm{t}$ and measuring the Euclidean distance to the detected sub-pixel corners.

Metric	Value
Total reprojection error ε(P)	10.8192 px^2
Mean reprojection error	0.3169 px
Maximum reprojection error	0.8772 px
RMSE	0.3506 px
Number of points evaluated	88 (8 x 11 grid)

The mean error of $0.3169\mathrm{px}$ is well below one pixel, indicating a highly accurate calibration. A visual overlay (figure 2) of measured corners (green) and reprojected points (red) confirms that the two sets are virtually indistinguishable at normal zoom.

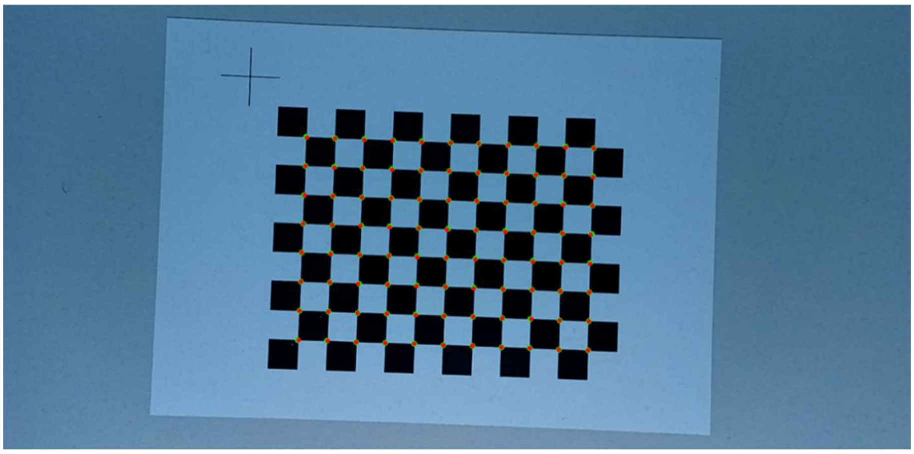

Figure 2

4.3 Cylinder Superimposition

The cylinder is rendered by drawing the projected base circle (red), top circle (blue), and lateral edges (green) as line segments. The 25 annotated images are arranged into a 5x5 mosaic. The visual inspection (figure 3) confirms that:

- The cylinder base is always correctly set to the chessboard plane ($Z = 0$), never floating above it.
- The cylinder shape and orientation change correctly with viewpoint.
- The cylinder scales consistently with the apparent size of the chessboard squares, indicating that the metric scale (mm) is preserved across all images.

These observations confirm that the estimated extrinsic matrices (R, t) correctly track the camera pose across different viewpoints.

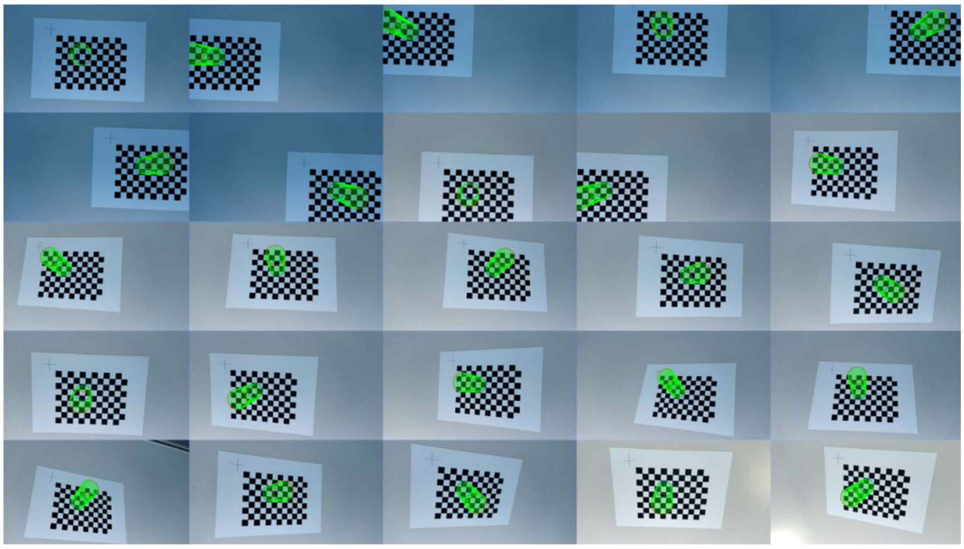

Figure 3

4.4 Principal Point Stability (Optimal Task)

To determine how many calibration images are needed for a stable estimate of the principal point, all possible subsets of size ℓ (ℓ = 2, 3, ..., 20) were enumerated from a fixed set of M = 20 images. For each subset, K was re-estimated using Zhang's linear method and the values of u_0 and v_0 were recorded. Subsets where B is not positive definite (Cholesky factorization fails) are discarded. The standard deviation of u_0 and v_0 was then computed across all valid subsets for each value of ℓ. Key observations from the resulting plot (figure 4):

- With $\ell = 2$ images the standard deviation is extremely high ($\mathrm{std}(c_x) = 916.50\mathrm{px}$, $\mathrm{std}(c_y) = 437.59\mathrm{px}$), indicating a completely unreliable estimate.
- Around ℓ = 6 images the standard deviation drops below 100px, showing a rapid improvement.
- By ℓ = 10 images, $\mathrm{std}(c_x) = 8.81$ px and $\mathrm{std}(c_y) = 7.99$ px.
- By ℓ = 15 images, $\mathrm{std}(c_x) = 4.48\mathrm{px}$ and $\mathrm{std}(c_y) = 3.40$ px, acceptable for most applications.
- At ℓ = 20 (all images), std = 0 since only one subset exists.

Taking about 10-12 images from different angles is sufficient to obtain a reliable calibration for this camera and chessboard.

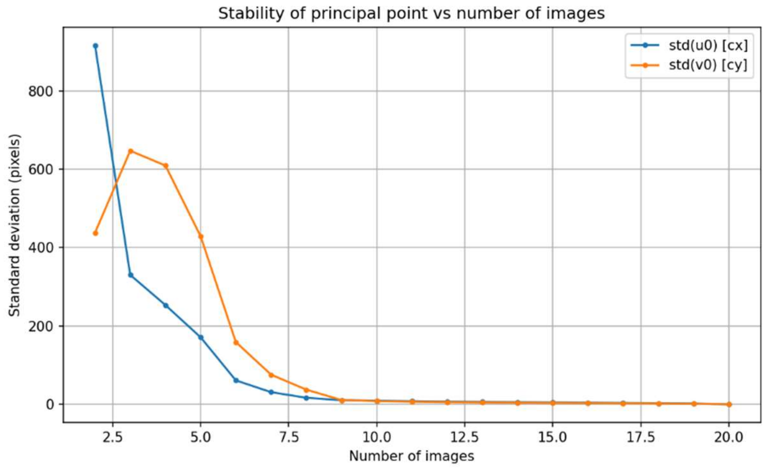

Figure 4

As an optional extension, the stability of the focal length $\mathrm{f} = \frac{(f_x + f_y)}{2}$ was also analysed. At ℓ = 2, $\mathrm{std}(f) = 2273.38\mathrm{px}$, which is much higher than for the principal point, indicating high instability with very few images. As more images are added, $\mathrm{std}(f)$ decreases consistently, reaching 46.20 px at ℓ = 10 and 17.65 px at ℓ = 15. The trend is consistent with the principal point behavior (figure 5).

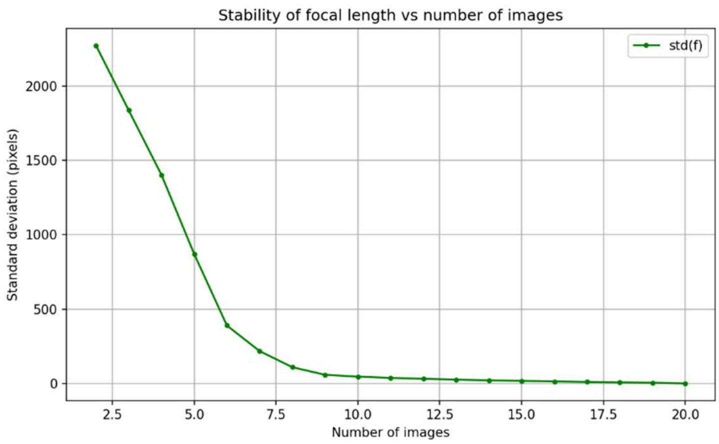

Figure 5

5. Discussion

These results clearly demonstrate that:

- Camera calibration using very few images (n < 5) is highly unstable.

- Increasing the number of images significantly improves the robustness the principal point estimate.
- Beyond approximately 10-12 images, additional images provide diminishing returns as the estimates have already converged.

6. Conclusion

In this project, a complete camera calibration pipeline was implemented from first principles, following Zhang's method without relying on built-in calibration libraries. Starting from 81 planar checkerboard images, all stages of the calibration process were explicitly derived, implemented, and validated.

Planar homographies were estimated using DLT and verified visually by projecting shapes onto the calibration images. The intrinsic camera matrix was recovered by solving Zhang's linear constraints, yielding physically plausible parameter ($\alpha_x = 1010.00\text{px}$, $\alpha_y = 1010.61\text{px}$, $\mathrm{s} = 1.69\text{px}$, $u_0 = 630.35\text{px}$, $v_0 = 373.90\text{px}$). Extrinsic parameters were computed analytically, and the rotation matrices were orthonormalised via SVD.

The calibration accuracy was quantified through the reprojection error: mean 0.3169px on the selected image, well below one pixel. A virtual cylinder was correctly superimposed on 25 images, providing qualitative validation of both intrinsics and extrinsics across different viewpoints.

Finally, the stability analysis showed that the standard deviation of the principal point decreases rapidly with the number of images, stabilising after approximately 10-12 views. This confirms the theoretical expectations of Zhang's method and the practical importance of using multiple images with sufficient geometric diversity.

7. References

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6. Anthropic, Claude, AI-based assistant used for debugging code, improving English expression, providing clarifications during the project and helping draft some parts of the report.